

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_diversity_weighted_replay.
- Best TSPLIB holdout gap: phase6_diversity_weighted_replay.
- Best synthetic holdout gap: phase6_failure_replay.

Run Metadata

- run_name: run_20260519_023137_m
- started_at_local: 2026-05-19 02:31:37
- finished_at_local: 2026-05-19 02:51:03
- duration_hhmm: 00:19
- duration_seconds: 1165.295
- seed_offset: 12000
- replicate_label: m
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.10708	0.003941	0.069575	0.236772	0.185832	0.0	0.794158	0.76	0.056868
phase6_random_replay	random	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.19457	0.857086	0.56	-0.029058
phase6_failure_replay	failure	score_only	False	0.121787	0.0	0.077501	0.171206	0.23931	0.221051	0.648022	0.76	0.059623

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.213387	0.064916	0.159398	0.236772	0.214847	0.19457	0.0	0.56	0.0
phase6_stratified_random_replay	stratified_random	score_only	False	0.122153	0.001628	0.078326	0.236772	0.156985	0.221162	0.850377	0.76	0.062297
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.076002	0.006169	0.050608	0.236772	0.161517	0.130039	0.760188	0.76	0.042501
phase6_residual_failure_replay	residual_failure	score_only	False	0.213387	0.064916	0.159398	0.149274	0.338865	0.291666	0.886718	0.56	-0.028087
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.140392	0.001628	0.089932	0.171206	0.258194	0.17235	0.787879	0.76	0.051766
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.213387	0.064916	0.159398	0.171206	0.348017	0.17024	0.0	0.56	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.10708, synthetic 0.003941, combined 0.069575.
- Accepted-epoch count 3, mean accepted code novelty 0.794158, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.185832, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency 0.056868 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.10708 across 7 instances; family means: ch=0.074431, kroD=0.176951, pcb=0.233979, pr=0.054466, rd=0.041972, st=0.093333.
- Panel synthetic_holdout mean gap 0.003941 across 4 instances; family means: clustered_gaussian=0.015764, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3447, "expected_gap": 0.406514, "family": "a", "name": "a280", "optimality_gap": 0.336565, "residual_gap": -0.069949}, {"best_known_cost": 629, "cost": 766, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.217806, "residual_gap": -0.039746}],

```
{"best_known_cost": 14379, "cost": 17442, "expected_gap": 0.328841, "family": "lin", "name": "lin105",  
"optimality_gap": 0.213019, "residual_gap": -0.115822}
```

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.857086, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133793, mean selected residual gap 0.037573.
- Adaptation efficiency -0.029058 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.406204, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.004886}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.134817}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.328006, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.000807}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.121787, synthetic 0.0, combined 0.077501.
- Accepted-epoch count 2, mean accepted code novelty 0.648022, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.23931, size bias 0.120322, and failure concentration 0.221051.
- Replay selection diversity 0.188305, mean selected expected gap 0.32049, mean selected residual gap -0.03642.
- Adaptation efficiency 0.059623 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 8, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.121787 across 7 instances; family means: ch=0.125157, kroD=0.102048, pcb=0.264898, pr=0.036113, rd=0.147282, st=0.051852.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 18716, "expected_gap": 0.326657, "family": "lin", "name": "lin105", "optimality_gap": 0.30162, "residual_gap": -0.025037}, {"best_known_cost": 629, "cost": 791, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.257552, "residual_gap": 0.002862}, {"best_known_cost": 2579, "cost": 3180, "expected_gap": 0.408608, "family": "a", "name": "a280", "optimality_gap": 0.233036, "residual_gap": -0.175572}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132582, mean selected residual gap 0.038784.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.409771, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.008453}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.137679}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.326657, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.002156}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.122153, synthetic 0.001628, combined 0.078326.
- Accepted-epoch count 2, mean accepted code novelty 0.850377, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.156985, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.237011, mean selected residual gap -0.017589.
- Adaptation efficiency 0.062297 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.122153 across 7 instances; family means: ch=0.142714, kroD=0.079224, pcb=0.184292, pr=0.099049, rd=0.101896, st=0.105185.
- Panel synthetic_holdout mean gap 0.001628 across 4 instances; family means: clustered_gaussian=0.006511, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3520, "expected_gap": 0.404343, "family": "a", "name": "a280", "optimality_gap": 0.36487, "residual_gap": -0.039473}, {"best_known_cost": 14379, "cost": 19404, "expected_gap": 0.324988, "family": "lin", "name": "lin105", "optimality_gap": 0.349468, "residual_gap": 0.02448}, {"best_known_cost": 629, "cost": 823, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.308426, "residual_gap": 0.052146}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.076002, synthetic 0.006169, combined 0.050608.
- Accepted-epoch count 3, mean accepted code novelty 0.760188, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.161517, size bias 8.3e-05, and failure concentration 0.130039.
- Replay selection diversity 0.263651, mean selected expected gap 0.167812, mean selected residual gap -0.020868.

- Adaptation efficiency 0.042501 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.076002 across 7 instances; family means: ch=0.055566, kroD=0.097962, pcb=0.168341, pr=0.056565, rd=0.041719, st=0.056296.
- Panel synthetic_holdout mean gap 0.006169 across 4 instances; family means: clustered_gaussian=0.024674, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 20125, "expected_gap": 0.325433, "family": "lin", "name": "lin105", "optimality_gap": 0.399611, "residual_gap": 0.074178}, {"best_known_cost": 2579, "cost": 3530, "expected_gap": 0.40318, "family": "a", "name": "a280", "optimality_gap": 0.368748, "residual_gap": -0.034432}, {"best_known_cost": 629, "cost": 784, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.246423, "residual_gap": -0.007631}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.886718, and final complexity 0.56.
- Active replay archive residual_archive with mean diversity 0.149274, mean hardness 0.338865, size bias 0.093558, and failure concentration 0.291666.
- Replay selection diversity 0.177489, mean selected expected gap 0.267129, mean selected residual gap 0.085263.
- Adaptation efficiency -0.028087 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.402947, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.001629}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.129094}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.323861, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.004952}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.140392, synthetic 0.001628, combined 0.089932.
- Accepted-epoch count 2, mean accepted code novelty 0.787879, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.258194, size bias 0.120322, and failure concentration 0.17235.
- Replay selection diversity 0.201683, mean selected expected gap 0.306411, mean selected residual gap -0.048111.
- Adaptation efficiency 0.051766 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.140392 across 7 instances; family means: ch=0.109168, kroD=0.125106, pcb=0.24164, pr=0.160338, rd=0.143995, st=0.093333.
- Panel synthetic_holdout mean gap 0.001628 across 4 instances; family means: clustered_gaussian=0.006511, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3333, "expected_gap": 0.402171, "family": "a", "name": "a280", "optimality_gap": 0.292361, "residual_gap": -0.10981}, {"best_known_cost": 14379, "cost": 18147, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.262049, "residual_gap": -0.059462}, {"best_known_cost": 629, "cost": 760, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.208267, "residual_gap": -0.058824}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.17024.
- Replay selection diversity 0.199284, mean selected expected gap 0.301724, mean selected residual gap 0.048265.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.402171, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.000853}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.127186}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.326838, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.001975}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Analysis

Main Evidence: Held-Out TSPLIB and Synthetic Holdout Gaps (Optimality and Transfer)

Condition	Replay Type	Mean Synthetic Gap	Mean Transfer Gap	Mean TSPLIB Gap	Transfer Rank (Lower Better)	TSPLIB Rank (Lower Better)
phase6_diversity_weighted_replay	Diversity-weighted Replay	0.006169	**0.050608**	**0.076002**	**1**	**1**
phase6_failure_replay	Failure Replay	0.0	0.077501	0.121787	3	3
phase6_stratified_random_replay	Stratified Random Replay	0.001628	0.078326	0.122153	4	4
phase6_diversity_failure_replay	Diversity + Failure Replay	0.001628	0.089932	0.140392	5	5

Condition	Replay Type	Mean Synthetic Gap	Mean Transfer Gap	Mean TSPLIB Gap	Transfer Rank (Lower Better)	TSPLIB Rank (Lower Better)
phase6_no_replay	No Replay	0.003941	0.069575	0.10708	2	2
phase6_random_replay	Random Replay	0.064916	0.159398	0.213387	8	8
phase6_residual_failure_replay	Residual Failure Replay	0.064916	0.159398	0.213387	8	8
phase6_random_replay_compression	Random Replay + Compression	0.064916	0.159398	0.213387	8	8
phase6_diversity_failure_replay_compression	Diversity + Failure Replay + Compression	0.064916	0.159398	0.213387	8	8

- **phase6_diversity_weighted_replay** shows the best transfer (lowest gaps on both held-out TSPLIB and synthetic holdout) and best held-out TSPLIB optimality gap.
- Failure replay (phase6_failure_replay) improves synthetic holdout gap to zero but shows moderate transfer gap and TSPLIB gap, slightly worse than no replay condition.
- Random based replays and residual failure replay show significant degradation on transfer metrics and held-out performance.
- Compression-aware versions lose code novelty (mean_code_novelty=0) and do not improve transfer or gaps over their non-compressed counterparts.

Replay Archive Analysis (Diversity, Hardness, Size Bias, Replay Failure Concentration)

Condition	Archive Key	Archive Size	Description or Diversity	Mean Hardness	Size Bias	Replay Failure Concentration (mean/final)	Replay Selection Diversity	Final Behavior Profile
phase6_no_replay	experience_archive	30	0.236772	0.185832	~0.0	0.0 / 0.0	0.0	clustered_constructor
phase6_random_replay	experience_archive	31	0.236772	0.214847	~0.0	0.19457 / 0.16	0.262174	balanced
phase6_failure_replay	worst_archive	29	0.171206	0.23931	0.120322	0.221051 / 0.219955	0.188305	clustered_constructor
phase6_random_replay_compression	experience_archive	31	0.236772	0.214847	~0.0	0.19457 / 0.16	0.262174	balanced
phase6_stratified_random_replay	experience_archive	30	0.236772	0.156985	~0.0	0.221162 / 0.251701	0.187367	clustered_constructor
phase6_diversity_weighted_replay	experience_archive	28	0.236772	0.181434	~0.0	0.130039 / 0.102041	0.263651	clustered_constructor

Condition	Archive Key	Archive Size	Description or Diversity	Mean Hardness	Size Bias	Replay Failure Concentration (mean/final)	Replay Selection Diversity	Final Behavior Profile
phase6_residual_failure_replay	experience_archive	31	0.149653	0.33856	0.094391	0.17024 / 0.291666	0.177489	balanced
phase6_diversity_failure_replay	worst_archive	29	0.174718	0.258194	0.120322	0.17235 / 0.174603	0.201683	clustered_constructor
phase6_diversity_failure_replay_compression	worst_archive	31	0.174718	0.348017	0.120322	0.17024 / 0.170068	0.199284	balanced

- Replay diversity is highest for the best transfer condition (diversity_weighted_replay) and no_replay (both ~0.237).
- Hardness is generally higher for conditions selecting failure or residual failure archives (0.24–0.35) but the highest hardness coincides with poor transfer and gaps (residual_failure_replay with 0.338 or diversity_failure_replay_compression with 0.348).
- Size bias is near-zero in experience archive-based replay; failure archive conditions show positive size bias (~0.12).
- Failure concentration is low in no_replay (0.0), moderate in diversity_weighted_replay (~0.13), and higher (~0.17–0.25) for failure or random replay conditions.

Code Novelty and Complexity

Condition	Mean Code Novelty	Last Code Novelty	Mean Complexity
phase6_diversity_weighted_replay	0.760188	0.662355	0.76
phase6_failure_replay	0.648022	0.648022	0.76
phase6_stratified_random_replay	0.850377	0.850377	0.76
phase6_no_replay	0.794158	0.853692	0.76
phase6_random_replay	0.857086	0.830563	0.56
phase6_random_replay_compression	0.0	0.0	0.56
phase6_residual_failure_replay	0.0	0.648022	0.56
phase6_diversity_failure_replay	0.787879	0.787879	0.76
phase6_diversity_failure_replay_compression	0.0	0.0	0.56

- Compression-aware conditions (random replay + compression, diversity failure replay + compression) show zero code novelty with no transfer improvement.
- The best transfer condition (diversity_weighted_replay) maintains high code novelty and complexity.
- Random replay conditions have lower complexity (0.56) than others (mostly 0.76).

Replay Mechanism Insights

- **Diversity-Weighted Replay** balances high archive diversity (~0.237), low failure concentration (~0.13), and moderate hardness (~0.18), achieving highest transfer and best TSPLIB optimality.
- **Failure Replay** concentrates on failure cases with higher hardness (0.24) and size bias (~0.12) but achieves zero synthetic gap yet moderate transfer and TSPLIB gaps.
- **Random Replay** with or without compression has higher failure concentration (~0.19) and selection diversity (~0.26), but poorest transfer and TSPLIB gaps.
- **Residual Failure Replay** has low diversity (~0.15), highest hardness (0.34), and high failure concentration (~0.17–0.29), but worst gaps, indicating overfitting or narrow focus.
- Compression pressures eliminate code novelty without transfer benefits.

Final Takeaways

- **phase6_diversity_weighted_replay** is the best transfer condition, showing the smallest gaps on held-out TSPLIB (0.076) and synthetic holdout (0.006) with low transfer gap (0.051) and highest code novelty among top performers. Archive diversity and moderate hardness support balanced learning.
- The **phase6_failure_replay** condition excels in synthetic holdout (0 gap), but has lower transfer and TSPLIB generalization than diversity-weighted replay.
- Random and residual failure replays have worse transfer performance and held-out optimality gaps despite some having higher failure concentration and selection diversity.
- Compression-aware replay variants sacrifice code novelty without improving transfer, suggesting a detrimental effect on learning.
- Code novelty positively correlates with transfer performance; conditions with zero novelty (compression) perform worse.
- Hardness and size bias alone do not guarantee transfer; balanced diversity and failure concentration are important.
- No replay performs reasonably but inferior to diversity-weighted replay in transfer gaps and TSPLIB performance.
- Lexical novelty (code novelty) is maintained in successful replay strategies but not in compression-aware runs.
- There is no evidence that residual failure replay or compressed replay produces superior transfer despite their algorithmic difference.

Conclusion

- Prioritize **phase6_diversity_weighted_replay** for best transfer generalization, balancing replay archive diversity, failure concentration, and code novelty.
- Failure replay improves synthetic holdout optimality gap but not overall transfer relative to diversity-weighted replay.
- Compression-aware replay reduces code novelty with no transfer gain.
- Failure concentration and hardness should be balanced; high value does not ensure better transfer.
- Random replay provides least transfer benefit.
- Lexical novelty corresponds here with transfer, but no isolated algorithmic invention from code novelty alone is indicated.

This conservative interpretation endorses **diversity-weighted replay** as the optimal replay mechanism for broad transfer in this benchmark suite.